

Exercise week 47-48

November 17-28, 2025 Kjersti Stangeland

Date: **Deadline is Friday November 28 at midnight**

Overarching aims of the exercises this week

The exercise set this week is meant as a summary of many of the central elements in various machine learning algorithms we have discussed through the semester. You don't need to answer all questions.

Linear and logistic regression methods

Question 1:

Which of the following is not an assumption of ordinary least squares linear regression?

- There is a linearity between predictors/features and target/output
- The inputs/features distributed according to a normal/gaussian distribution

Answer

In OLS we assume there is linearity between features and target - that's the whole essence of it. We do not assume that the features are distributed according to a normal distribution.

Question 2:

The mean squared error cost function for linear regression is convex in the parameters, guaranteeing a unique global minimum. True or False? Motivate your answer.

Answer True. The cost function is a quadratic function in the parameters, and such a function is only convex if the Hessian matrix is positive semidefinite. Meaning the

matrix $X^T X$ is positive semidefinite as $v^T (X^T X) v = (Xv)^T (Xv) = \|Xv\|^2 \geq 0$

Question 3:

Which statement about logistic regression is false?

- Logistic regression is used for binary classification.
- It uses the sigmoid function to map linear scores to probabilities.
- It has an analytical closed-form solution.
- Its log-loss (cross-entropy) is convex.

Answer It does not have an analytical closed-form solution.

Question 4:

Logistic regression produces a linear decision boundary in the input space. True or False? Explain.

Answer Yes, it draws a "straight line", which defines two cases, say yes or no.

Question 5:

Give two reasons why logistic regression is preferred over linear regression for binary classification.

Answer Logistic regression is preferred over linear regression for binary classification as it can give probability outputs (value between 0 and 1) and is designed for classification as it uses cross-entropy as loss function (not MSE which measures continuous prediction).

Neural networks

Question 6:

Which statement is not true for fully-connected neural networks?

- Without nonlinear activation functions they reduce to a single linear model.
- Training relies on backpropagation using the chain rule.
- A single hidden layer can approximate any continuous function on a compact set.

- The loss surface of a deep neural network is convex.

Answer "The loss surface of a deep neural network is convex" is false.

Question 7:

Using sigmoid activations in many layers of a deep neural network can cause vanishing gradients. True or False? Explain.

Answer True! Sigmoid outputs from 0 to 1. If the inputs are very small or very large for many neurons, the gradient become small and the learning gets slow.

Question 8:

Describe the vanishing gradient problem: Why does it occur? Mention one technique to mitigate it and explain briefly.

Answer Hehe already did, but the training is done by updating the weights using the gradients. If the gradients are small, learning becomes slow. To mitigate one can use ReLU og LReLU as activation functions. These have deriavtives of 1 for positive inputs, so the gradients arent shrinked repeatedly.

Question 9:

Consider a fully-connected network with layer sizes n_0 (the input layer) , n_1 (first hidden layer) , $\dots$, n_L , where n_L is the output layer. Derive a general formula for the total number of trainable parameters (weights + biases).

Answer

Each layer has a weight matrix and a bias vector, which align with the size of the layer. So to get to the number of trainable parameters, we have to sum them over the number of layers. Every neuron in layer l connects to all neurons in the previous layer $l - 1$, in addition each neuron in layer l has a corresponding bias. So for layer l , the weight matrix is of size (n_l, n_{l-1}) and the bias is $(n_l, 1)$.

So in total:

$$\text{Number of parameters} = \sum_{l=1}^L (n_l \cdot n_{l-1} + n_l)$$

Convolutional Neural Networks

Question 10:

Which of the following is not a typical property or advantage of CNNs?

- Local receptive fields
- Weight sharing
- More parameters than fully-connected layers
- Pooling layers offering some translation invariance

Answer

Question 11:

Using zero-padding in convolutional layers can preserve the input spatial dimensions when using a 3×3 kernel/filter, stride 1, and padding $P = 1$. True or False?

Question 12:

Given input width W , kernel size K , stride S , and padding P , derive the formula for the output width $W_{\text{out}} = \frac{W-K+2P}{S} + 1$.

Question 13:

A convolutional layer has: C_{in} input channels, C_{out} output channels (filters) and kernel size $K_h \times K_w$. Compute the number of trainable parameters including biases.

Recurrent Neural Networks

Question 14:

Which statement about simple RNNs is false?

- They maintain a hidden state updated each time step.
- They use the same weight matrices at every time step.
- They handle sequences of arbitrary length.
- They eliminate the vanishing gradient problem.

Question 15:

LSTMs mitigate the vanishing gradient problem by using gating mechanisms (input, forget, output gates). True or False? Explain.

Question 16:

What is Backpropagation Through Time (BPTT) and why is it required for training RNNs?

Question 17:

What does a sliding window do? And why would we use it?